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(54) **COMPOSITE TEXTILE**

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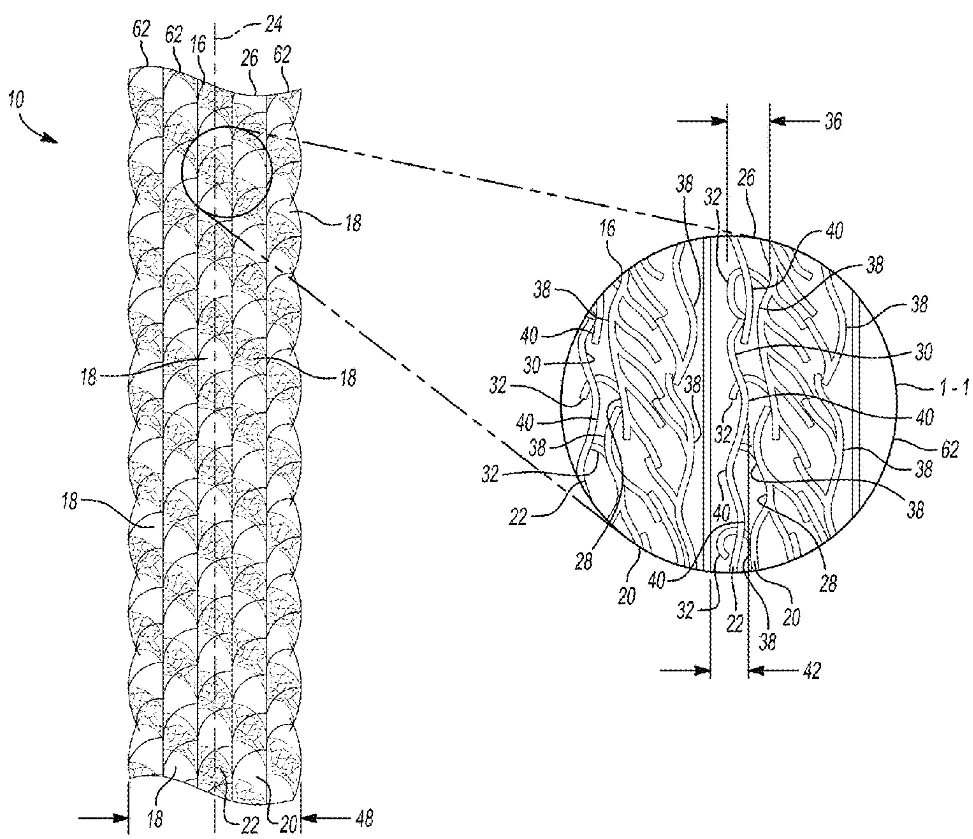
(57) **ABSTRACT**

A composite textile includes a first layer formed from a two-yarn composite that includes a first yarn formed from meta-aramid fibers and a second yarn formed from polyester fibers. The composite textile includes a second layer formed from the two-yarn composite and abutting the first layer along a longitudinal axis. The first yarn has a first surface, the second yarn has a second surface, and the first yarn includes hooks each extending from the first surface toward the second surface. The first layer includes a first plurality of micro-ridges disposed along the first surface and the second layer includes a second plurality of micro-ridges disposed along the second surface. The first and second plurality of micro-ridges interact to generate friction between the first and second layers and limit movement of the first and second layers with respect to one another as the textile extends along the longitudinal axis.



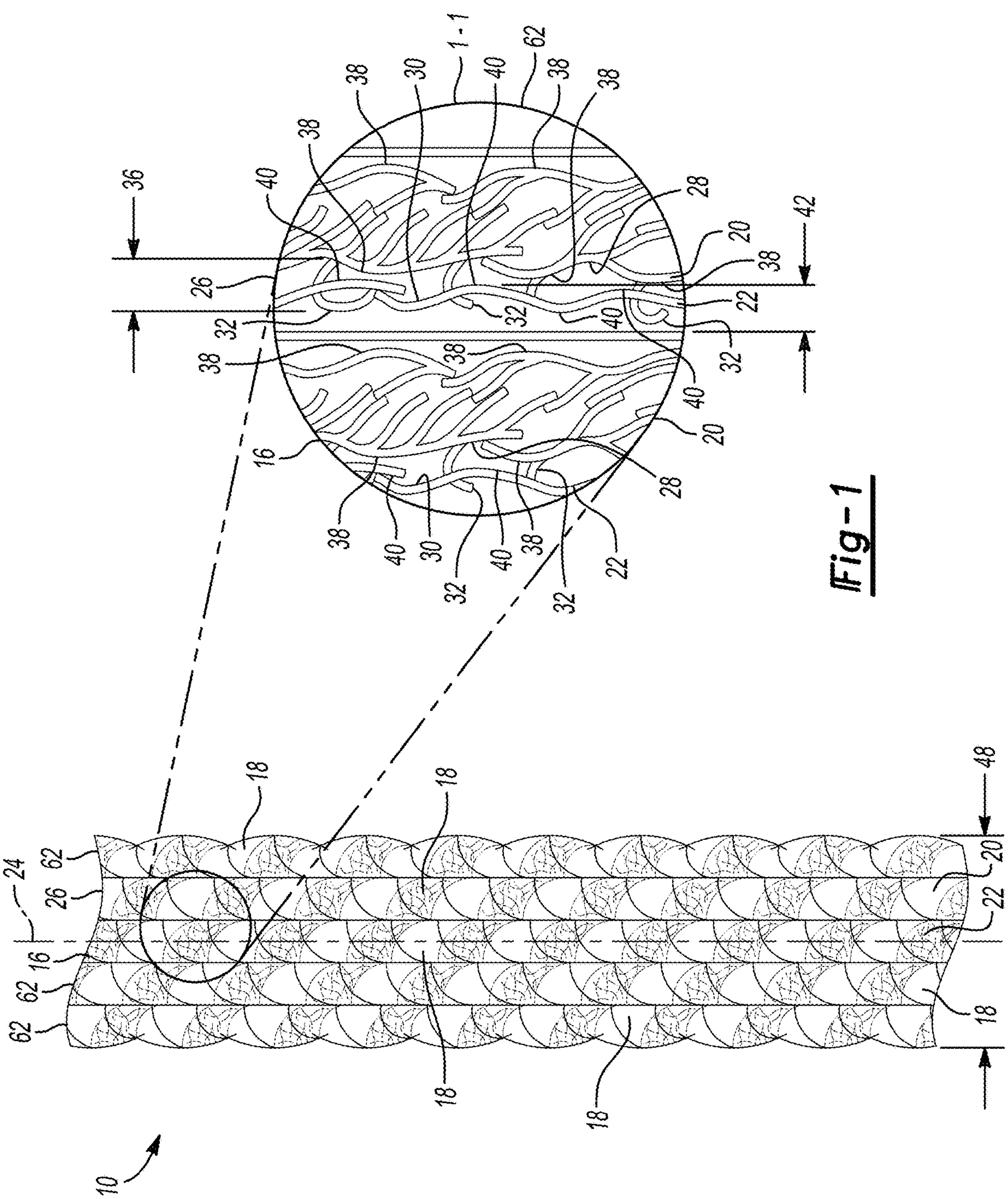


Fig-1

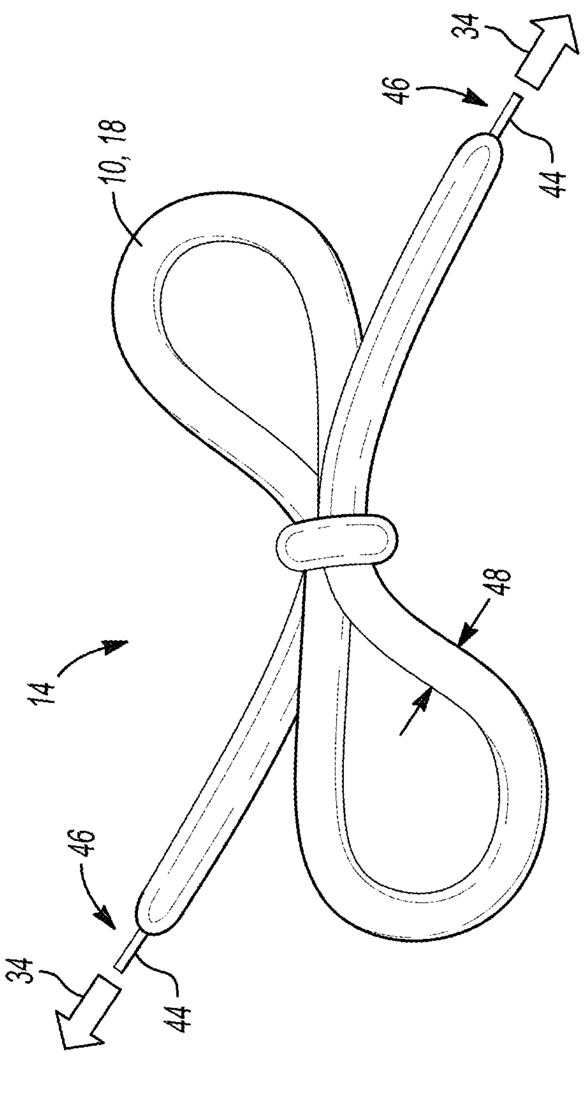


Fig-2

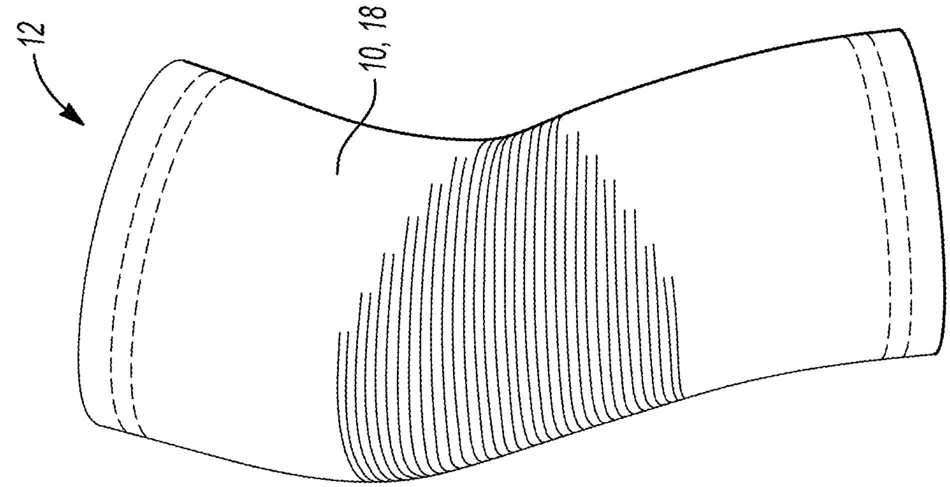


Fig-3

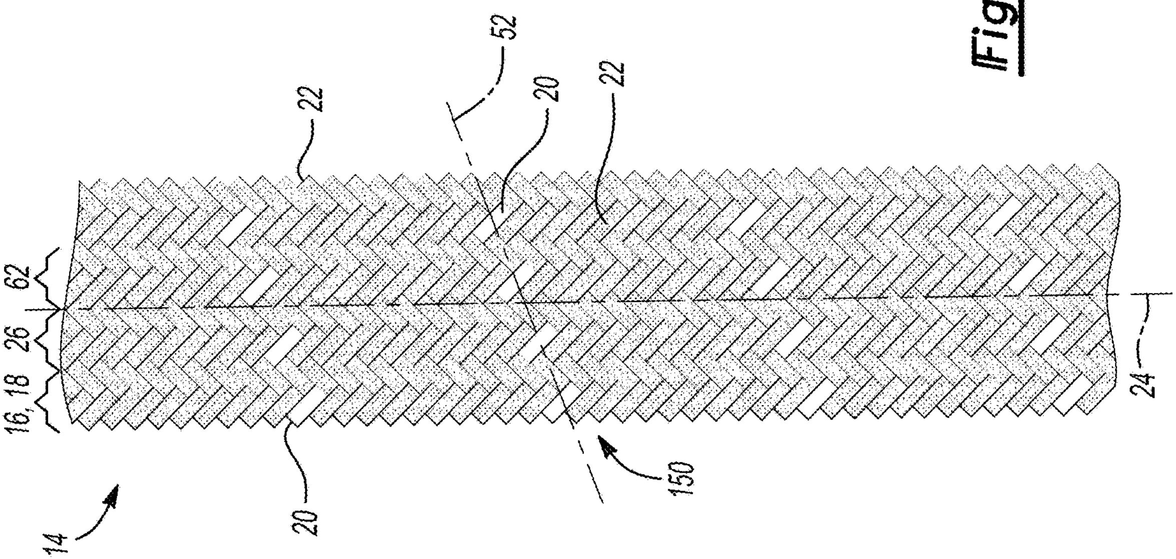


Fig-5

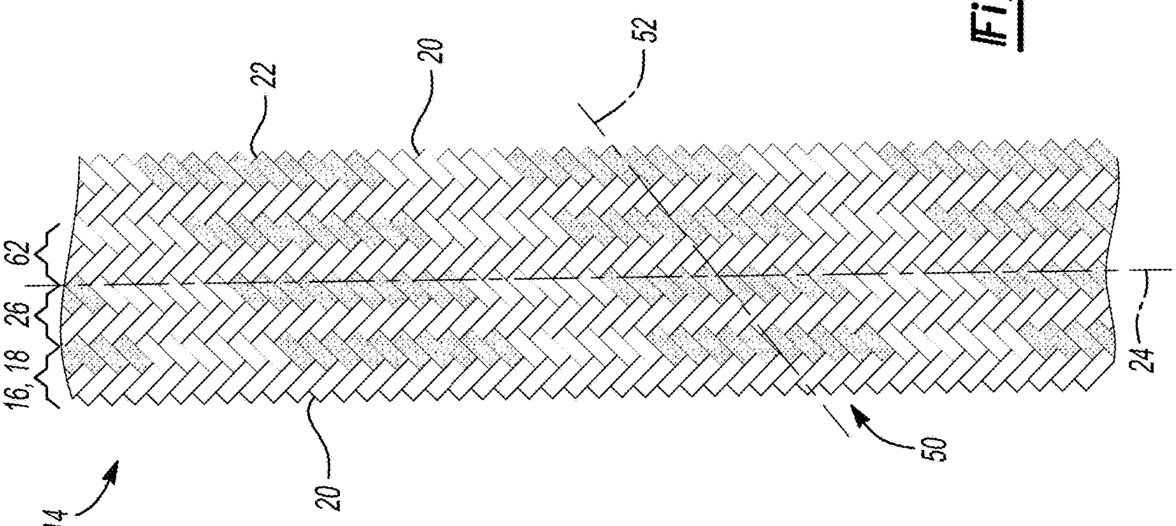


Fig-4

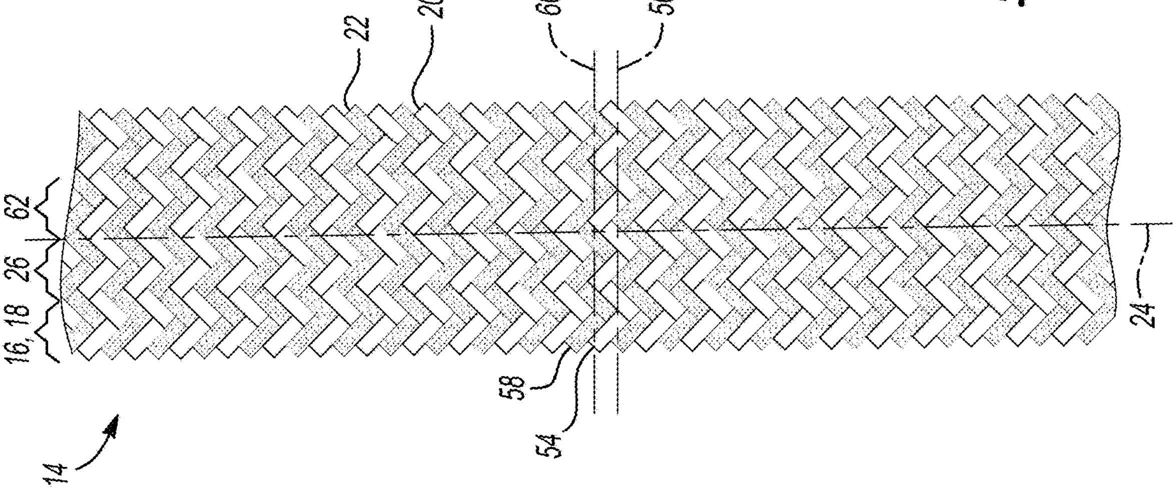


Fig-7

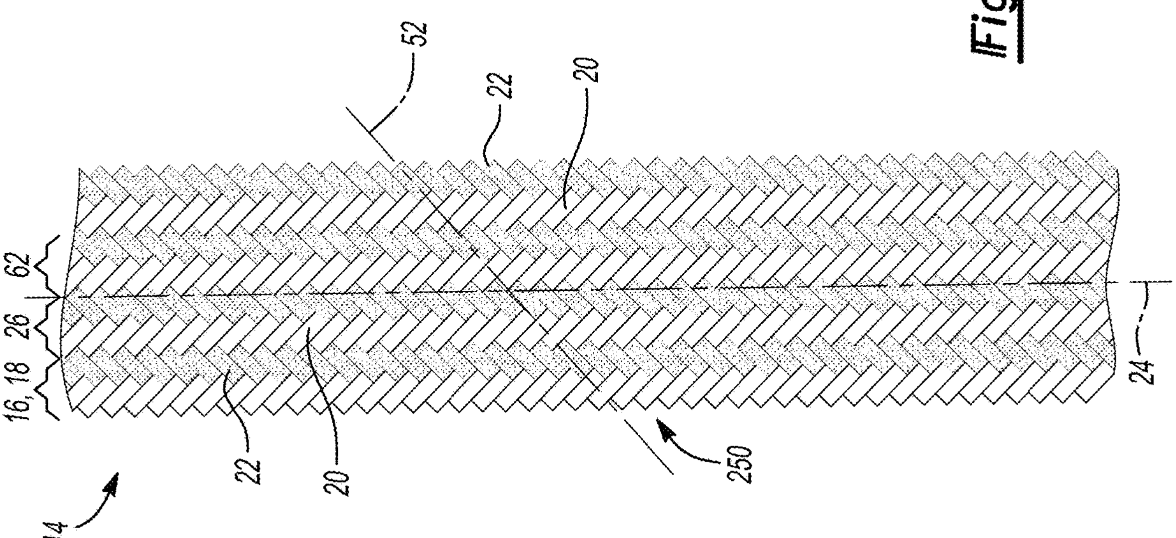


Fig-6

COMPOSITE TEXTILE

CROSS-REFERENCE TO RELATED APPLICATIONS

[0001] This application claims the benefit of priority to U.S. Provisional Patent Application No. 63/563,358, filed on Mar. 9, 2024, which is hereby incorporated by reference in its entirety.

INTRODUCTION

[0002] The disclosure relates to a composite textile.

[0003] Textiles, such as fabrics and other fiber-based materials, may be formed from natural and/or synthetic fibers and may be useful for apparel and equipment applications requiring both comfort and functionality. For example, high-performance athletic equipment formed from textiles may be used in harsh environments in which the textile is exposed to heat, tensile forces, and strain. Such applications may therefore require textiles having excellent appearance, breathability, strength, elasticity, grip, and durability. Under certain conditions, such textiles may also experience changes in shape and mechanical properties. For example, textile shoelaces may stretch, come untied, or loosen during athletic activities.

SUMMARY

[0004] A composite textile includes a first layer formed from a two-yarn composite that includes a first yarn formed from a plurality of meta-aramid fibers and a second yarn formed from a plurality of polyester fibers. The first layer extends along a longitudinal axis. The composite textile also includes a second layer formed from the two-yarn composite and abutting the first layer along the longitudinal axis. The first yarn has a first surface having a first surface finish, and the second yarn has a second surface spaced opposite the first surface and having a second surface finish that is smoother than the first surface finish. The first yarn includes a plurality of hooks each extending from the first surface toward the second surface. The first layer includes a first plurality of micro-ridges disposed along the first surface and the second layer includes a second plurality of micro-ridges disposed along the second surface. The first plurality of micro-ridges and the second plurality of micro-ridges interact to generate friction between the first layer and the second layer and limit movement of the first layer and the second layer with respect to one another as the composite textile extends along the longitudinal axis in response to a tensile force.

[0005] In one aspect, the plurality of meta-aramid fibers may be present in the composite textile in a quantity of from 30 parts to 95 parts based on 100 parts of the composite textile.

[0006] In an additional aspect, the plurality of polyester fibers may be present in the composite textile in an amount of from 5 parts to 70 parts based on 100 parts of the composite textile.

[0007] In another aspect, the first layer and the second layer may align along the longitudinal axis such that the first plurality of micro-ridges and the second plurality of micro-ridges contact one another when one of the first layer and the second layer moves along the longitudinal axis with respect to another of the first layer and the second layer in response to the tensile force.

[0008] In a further aspect, the plurality of hooks may attach to the second surface along the longitudinal axis to limit movement of one of the first layer and the second layer with respect to another of the first layer and the second layer along the longitudinal axis in response to the tensile force.

[0009] In one aspect, each of the plurality of hooks may have a height of from 1 micron to 50 microns.

[0010] In an additional aspect, the first yarn may be braided with the second yarn about the longitudinal axis.

[0011] In another aspect, the first yarn may be woven with the second yarn about the longitudinal axis.

[0012] In a further aspect, the first yarn may be knitted with the second yarn about the longitudinal axis.

[0013] In one aspect, the composite article may further include at least one additional layer formed from the two-yarn composite and abutting the second layer along the longitudinal axis to therefore sandwich the second layer between the first layer and the at least one additional layer.

[0014] In an additional aspect, the first yarn may have a yarn count of 21 meta-aramid fibers and a single ply, and the second yarn may be a spun yarn.

[0015] In another aspect, a shoelace may be formed from the composite textile and may include an aglet disposed at each end of the shoelace.

[0016] In a further aspect, the shoelace may have a width of from 3 millimeters (mm) to 30 mm.

[0017] In one aspect, a fabric may be formed from the composite textile.

[0018] In another embodiment, a composite textile includes a first layer formed from a two-yarn composite that includes a first yarn formed from a plurality of meta-aramid fibers and a second yarn formed from a plurality of polyester fibers. The first layer extends along a longitudinal axis. The composite textile also includes a second layer formed from the two-yarn composite and abutting the first layer along the longitudinal axis. The composite textile further includes a third layer formed from the two-yarn composite and abutting the second layer along the longitudinal axis to thereby sandwich the second layer between the first layer and the second layer. The plurality of meta-aramid fibers and the plurality of polyester fibers are present in the composite textile in a ratio of a quantity of the plurality of meta-aramid fibers to an amount of the plurality of polyester fibers of from 5:95 to 95:5. The first yarn has a first surface and the second yarn has a second surface spaced opposite the first surface, and the first yarn includes a plurality of hooks each extending from the first surface toward the second surface. The first layer includes a first plurality of micro-ridges disposed along the first surface and the second layer includes a second plurality of micro-ridges disposed along the second surface. The first plurality of micro-ridges and the second plurality of micro-ridges interact to generate friction between the first layer and the second layer and limit movement of the first layer and the second layer with respect to one another as the composite textile extends along the longitudinal axis in response to a tensile force.

[0019] In one aspect, the first layer and the second layer may include the first yarn arranged in a first zig-zag pattern along a first horizontal axis that is perpendicular to the longitudinal axis. The first layer and the second layer may include the second yarn arranged in a second zig-zag pattern that abuts the first zig-zag pattern along a second horizontal axis that is parallel to the first horizontal axis.

[0020] In an additional aspect, the first layer and the second layer may include the first yarn and the second yarn arranged in a staggered pattern along an axis that is oblique to the longitudinal axis.

[0021] In another aspect, the first layer and the second layer may align along the longitudinal axis such that the first plurality of micro-ridges and the second plurality of micro-ridges contact one another when the first layer translates along the longitudinal axis with respect to the second layer in response to the tensile force.

[0022] In a further aspect, the plurality of hooks may attach to the second surface along the longitudinal axis to limit translation of the first layer with respect to the second layer along the longitudinal axis in response to the tensile force.

[0023] In a further embodiment, a composite textile includes a first layer formed from a two-yarn composite that includes a first yarn formed from a plurality of meta-aramid fibers and a second yarn formed from a plurality of polyester fibers, wherein the first layer extends along a longitudinal axis. The composite textile also includes a second layer formed from the two-yarn composite and abutting the first layer along the longitudinal axis. The plurality of meta-aramid fibers and the plurality of polyester fibers are present in the composite textile in a ratio of a quantity of the plurality of meta-aramid fibers to an amount of the plurality of polyester fibers of from 5:95 to 95:5. The first yarn has a first surface having a first surface finish, a yarn count of 21 meta-aramid fibers, and a single ply. The second yarn is a spun yarn and has a second surface spaced opposite the first surface and having a second surface finish that is smoother than the first surface finish. The first yarn includes a plurality of hooks each extending from the first surface toward the second surface and having a height of from 1 micron to 50 microns. The first layer includes a first plurality of micro-ridges disposed along the first surface and the second layer includes a second plurality of micro-ridges disposed along the second surface. The first plurality of micro-ridges and the second plurality of micro-ridges contact one another to generate friction between the first layer and the second layer and limit movement of the first layer and the second layer with respect to one another as the composite textile extends along the longitudinal axis in response to a tensile force. The plurality of hooks attach to the second surface along the longitudinal axis to limit movement of one of the first layer and the second layer with respect to another of the first layer and the second layer along the longitudinal axis in response to the tensile force.

[0024] The above features and advantages, and other features and attendant advantages of this disclosure, will be readily apparent from the following detailed description of illustrative examples and modes for carrying out the present disclosure when taken in connection with the accompanying drawings and the appended claims. Moreover, this disclosure expressly includes combinations and sub-combinations of the elements and features presented above and below.

BRIEF DESCRIPTION OF THE DRAWINGS

[0025] FIG. **1** is a schematic illustration of a plan view of a composite textile including a two-yarn composite shown in a magnified portion at **1-1**.

[0026] FIG. **2** is a schematic illustration of a perspective view of a shoelace formed from the composite textile of FIG. **1**.

[0027] FIG. **3** is a schematic illustration of a perspective view of a fabric formed from the composite textile of FIG. **1**.

[0028] FIG. **4** is a schematic illustration of a plan view of a portion of one configuration of the shoelace of FIG. **2**, wherein the two-yarn composite is arranged in a staggered pattern.

[0029] FIG. **5** is a schematic illustration of a plan view of a portion of another configuration of the shoelace of FIG. **2**, wherein the two-yarn composite is arranged in another staggered pattern.

[0030] FIG. **6** is a schematic illustration of a plan view of a portion of yet another configuration of the shoelace of FIG. **2**, wherein the two-yarn composite is arranged in yet another staggered pattern.

[0031] FIG. **7** is a schematic illustration of a plan view of a portion of another configuration of the shoelace of FIG. **2**, wherein the two-yarn composite is arranged in a zig-zag pattern.

DETAILED DESCRIPTION

[0032] Referring to the Figures, wherein like reference numerals refer to like elements, a composite textile **10** (FIG. **1**) is shown generally. The composite textile **10** may be useful for forming, for example, athletic recovery, support, and fastening equipment having excellent aesthetics, durability, and comfort. In particular, the composite textile **10** may be useful for forming high-performance fabrics **12** (FIG. **3**) and shoelaces **14** (FIG. **2**) that require excellent tensile strength, elasticity, grip, flame retardance, performance, and reliability.

[0033] Referring to FIGS. **2** and **3**, the composite textile **10** may be useful for consumer products and sporting goods applications such as, but not limited to, shoelaces **14** (FIG. **2**); fabric **12** (FIG. **3**) for athletic braces such as knee, ankle, shoulder, elbow, wrist, and knee braces; athletic tapes; ankle ties or foot straps for rowing applications; gloves; sleeves; cloths; stitching; and the like. In one non-limiting example, the composite textile **10** may be useful for forming shoelaces **14** for industrial footwear applications such as safety boots, footwear for law enforcement, fire retardant footwear, and the like; for footwear worn while participating in sports such as tennis, basketball, soccer, football, track and field, cross-country running, and hockey; for footwear worn while participating in leisure activities such as walking, running, and hiking; and for footwear designed for children, elderly, and individuals with a disability. More specifically, shoelaces **14** formed from the composite textile **10** have excellent grip, resistance to loosening, i.e., untie-resistance, tensile strength, elasticity, cut-resistance, and flame-resistance, as set forth in more detail below.

[0034] Alternatively, the composite textile **10** may be useful for other non-sporting goods applications, such as, but not limited to, packaging materials, such as netting and twine; fastening materials, such as cables, straps, ropes, and lacing; tooling applications, such as cabling, stitching, and fasteners; industrial clothing, such as protective garments, aprons, reinforcement stitching, and industrial workwear; and medical devices, such as support garments, compression sleeves, and the like.

Composite Textile

[0035] Referring now to FIG. **1**, the composite textile **10** includes a first layer **16** formed from a two-yarn composite

18 that includes a first yarn **20** formed from a plurality of meta-aramid fibers and a second yarn **22** formed from a plurality of polyester fibers. That is, the first layer **16** is a composite formed from two yarns **20**, **22**. The first yarn **20** may have a yarn count of 21 meta-aramid fibers and a single ply and may be classified as 21 s/1 meta-aramid fiber yarn. The second yarn **22** may be a spun yarn and may be classified as a spun polyester yarn. Further, the first layer **16** formed from the two-yarn composite **18** extends along a longitudinal axis **24**. That is, the first layer **16** may extend lengthwise along an article.

[0036] Referring again to FIG. **1**, the composite textile **10** also includes a second layer **26** formed from the two-yarn composite **18** and abutting the first layer **16** along the longitudinal axis **24**. That is, the second layer **26** is formed from the two-yarn composite **18** that includes the first yarn **20** formed from the plurality of meta-aramid fibers and the second yarn **22** formed from the plurality of polyester fibers and contacts the first layer **16**. For example, the first yarn **20** may braided with the second yarn **22** about the longitudinal axis **24**. That is, three or more strands of the first yarn **20** and the second yarn **22** may be interwoven diagonally to create a strong, flexible cord or shoelace **14** (FIG. **2**) formed from the composite textile **10**.

[0037] In another non-limiting example, the first yarn **20** may be woven with the second yarn **22** about the longitudinal axis **24**. For example, fabric **12** (FIG. **3**) may be formed from the composite textile **10**. That is, the first yarn **20** and the second yarn **22** may be interlaced vertically (warp) and horizontally (weft) at right angles to create the fabric **12** (FIG. **3**) formed from the composite textile **10**. In another non-limiting example, the first yarn **20** may be knitted with the second yarn **22** about the longitudinal axis **24**. That is, the first yarn **20** and second yarn **22** may form loops in rows (weft) or columns (warp) to create the fabric **12** formed from the composite textile **10**.

[0038] Referring again to FIG. **1** and best shown at magnified portion **1-1**, the first yarn **20** has a first surface **28** having a first surface finish. Further, the second yarn **22** has a second surface **30** spaced opposite the first surface **28** and having a second surface finish that is smoother than the first surface finish. That is, the first yarn **20** may have a comparatively coarse surface finish and the second yarn **22** may have a comparatively smooth surface finish.

[0039] More specifically, as best shown at magnified portion **1-1** of FIG. **1**, the first yarn **20** includes a plurality of hooks **32** each extending from the first surface **28** toward the second surface **30**. The plurality of hooks **32** may attach to the second surface **30** along the longitudinal axis **24** to limit movement of one of the first layer **16** and the second layer **26** with respect to another one of the first layer **16** and the second layer **26** along the longitudinal axis **24** in response to a tensile force **34** (FIG. **2**). For example, the plurality of hooks **32** may attach to the second surface **30** along the longitudinal axis **24** to limit translation of the first layer **16** with respect to the second layer **26** along the longitudinal axis **24** in response to the tensile force **34**. That is, the plurality of hooks **32** may engage with the second surface **30** to anchor the first yarn **20** with respect to the second yarn **22** along the longitudinal axis **24** to thereby limit undesired stretching or elongation of the composite textile **10**. For embodiments in which the shoelace **14** (FIG. **2**) is formed from the composite textile **10**, such limited translation along the longitudinal axis **24** may contribute to an excellent resistance to undesired loosening or untying of the shoelace **14** during use and may enhance an ability of the shoelace to remain tied or fastened.

[0040] Each of the plurality of hooks **32** may be measured on a micron or micrometer scale, i.e., 1×10^{-6} meters. For example, each of the plurality of hooks **32** may have a height **36** (FIG. **1** at **1-1**) of from 1 micron to 50 microns, or from 5 microns to 45 microns, or from 10 microns to 40 microns, or from 20 microns to 30 microns. Hooks **32** having a height **36** that is less than 1 micron or greater than 50 microns may be unsuitable for extending towards or effectively contacting the second surface **30** to thereby limit translation along the longitudinal axis **24**.

[0041] As described with continued reference to magnified portion **1-1** of FIG. **1**, the first layer **16** also includes a first plurality of micro-ridges **38** disposed along the first surface **28**, and the second layer **26** includes a second plurality of micro-ridges **40** disposed along the second surface **30**. The first plurality of micro-ridges **38** and the second plurality of micro-ridges **40** interact to generate friction between the first layer **16** and the second layer **26** and limit movement of the first layer **16** and the second layer **26** with respect to one another as the composite textile **10** extends along the longitudinal axis **24** in response to the tensile force **34** (FIG. **2**). That is, the first layer **16** and the second layer **26** may align along the longitudinal axis **24** such that the first plurality of micro-ridges **38** and the second plurality of micro-ridges **40** contact one another when one of the first layer **16** and the second layer **26** moves along the longitudinal axis **24** with respect to one another of the first layer **16** and the second layer **26** in response to the tensile force **34**. For example, the first layer **16** and the second layer **26** may align along the longitudinal axis **24** such that the first plurality of micro-ridges **38** and the second plurality of micro-ridges **40** contact and minimize or prevent sliding when the first layer **16** translates or moves along the longitudinal axis **24** with respect to the second layer **26** in response to the tensile force **34**. That is, the first plurality of micro-ridges **38** and the second plurality of micro-ridges **40** may abut and interlock with one another to anchor the first yarn **20** with respect to the second yarn **22** along the longitudinal axis **24** and thereby limit undesired stretching or elongation of the composite textile **10**. For embodiments in which the shoelace **14** (FIG. **2**) is formed from the composite textile **10**, such limited translation along the longitudinal axis **24** may contribute to an excellent resistance to loosening of the shoelace **14** and may enhance an ability of the shoelace to remain tied or fastened during use.

[0042] Each of the first plurality of micro-ridges **38** and the second plurality of micro-ridges **40** may also be measured on a micron or micrometer scale. For example, each of the first plurality of micro-ridges **38** and the second plurality of micro-ridges **40** may have a height **42** (FIG. **1** at **1-1**) of from 1 micron to 50 microns, or from 5 microns to 45 microns, or from 10 microns to 40 microns, or from 20 microns to 30 microns. Micro-ridges **38**, **40** having a height **42** that is less than 1 micron may be unsuitable for interacting to generate friction between the first layer **16** and the second layer **26** to thereby limit translation along the longitudinal axis **24** in response to the tensile force **34**, and may instead slide over one another to permit translation along the longitudinal axis **24**. Likewise, micro-ridges **38**, **40** having a height **42** that is greater than 50 microns may interfere with one another and prevent a smooth appearance or other desired aesthetic of the shoelace **14**.

Fiber Ratios

[0043] Referring again to the two-yarn composite **18** of the composite textile **10**, the plurality of meta-aramid fibers may be present in the composite textile **10** in a quantity of from 30 parts to 95 parts, or from 35 parts to 90 parts, or from 40 parts to 80 parts, or from 55 parts to 70 parts, based on 100 parts of the composite textile **10**. Conversely, the plurality of polyester fibers may be present in the composite textile **10** in an amount of from 5 parts to 70 parts, or from 10 parts to 65 parts, or from 20 parts to 60 parts, or from 30 parts to 45 parts, based on 100 parts of the composite textile **10**. Stated differently, in some embodiments, the plurality of meta-aramid fibers and the plurality of polyester fibers are present in the composite textile **10** in a ratio or proportion of the quality of the plurality of meta-aramid fibers to the amount of the plurality of polyester fibers of from 5:95 to 95:5, or from 15:85 to 85:15, or from 20:80 to 80:20, or from 30:70 to 70:30, or from 40:60 to 60:40, or from 45:55 to 55:45.

[0044] Each of the aforementioned quantities, amounts, and ratios may be critical for specific applications and desired performance properties of the composite textile **10**, as set forth in more detail below. As such, textiles that include the plurality of meta-aramid fibers and the plurality of polyester fibers present in the composite textile **10** in a ratio or proportion of less than from 5:95 or greater than 95:5 may have unsuitable grip, be susceptible to loosening, have low tensile strength and elasticity, have an undesirable weight, and/or have low flame-resistance.

Shoelace Applications

[0045] As a non-limiting example described with reference to FIG. **2**, for certain applications, the shoelace **14** may be formed form the composite textile **10** and may include an aglet **44** disposed at each end **46** of the shoelace **14**. The shoelace **14** may have a width **48** of from 3 millimeters (mm) to 30 mm, or from 6 mm to 15 mm, or 7 mm, or 8 mm, depending on a desired use and application of the shoelace **14**. Further, the shoelace **14** may have a flat or round shape. At widths **48** of less than 3 mm or greater than 30 mm, the shoelace **14** may not exhibit adequate untie-resistance, tensile strength, elasticity, or flame-resistance.

[0046] As a non-limiting example, shoelaces **14** useful for lacing tennis footwear may have a width **48** of 8 mm and may include 80 parts of the plurality of meta-aramid fibers and 20 parts of the plurality of polyester fibers based on 100 parts of the composite textile **10**. At this fiber loading quantity and amount, the shoelace **14** may have comparatively very high untie-resistance, comparatively very high tensile strength, comparatively low elasticity, comparatively low flame-resistance, a comparatively moderately heavy weight, and a median melting temperature of about 50° C.

[0047] For shoelaces **14** useful for lacing basketball footwear, the shoelace **14** may be round, have a width **48** of 6 mm, and may include 55 parts of the plurality of meta-aramid fibers and 45 parts of the plurality of polyester fibers based on 100 parts of the composite textile **10**. At this fiber loading quantity and amount, the shoelace **14** may have comparatively moderately high untie-resistance, comparatively high tensile strength, comparatively high elasticity, comparatively high flame-resistance, a comparatively light weight, and a median melting temperature of about 110° C.

[0048] For shoelaces **14** useful for lacing soccer footwear, the shoelace **14** may have a width **48** of 6 mm and may include 80 parts of the plurality of meta-aramid fibers and 20 parts of the plurality of polyester fibers based on 100 parts of the composite textile **10**. At this fiber loading quantity and amount, the shoelace **14** may have comparatively very high untie-resistance, comparatively very high tensile strength, comparatively high elasticity, comparatively moderate flame-resistance, a comparatively very light weight, and a median melting temperature of about 50° C.

[0049] For shoelaces **14** useful for lacing American football footwear that is predominately useful for kicking a football, the shoelace **14** may have a width **48** of 6 mm and may include 80 parts of the plurality of meta-aramid fibers and 20 parts of the plurality of polyester fibers based on 100 parts of the composite textile **10**. At this fiber loading quantity and amount, the shoelace **14** may have comparatively very high untie-resistance, comparatively very high tensile strength, comparatively high elasticity, comparatively moderate flame-resistance, a comparatively very light weight, and a median melting temperature of about 50° C.

[0050] For shoelaces **14** useful for lacing American football footwear that is predominately useful for running, the shoelace **14** may have a width **48** of 7 mm and may include 40 parts of the plurality of meta-aramid fibers and 60 parts of the plurality of polyester fibers based on 100 parts of the composite textile **10**. At this fiber loading quantity and amount, the shoelace **14** may have comparatively very high untie-resistance, comparatively high tensile strength, comparatively low elasticity, comparatively high flame-resistance, a comparatively light weight, and a median melting temperature of about 150° C.

[0051] For shoelaces **14** useful for lacing running footwear, the shoelace **14** may have a width **48** of 7 mm and may include 40 parts of the plurality of meta-aramid fibers and 60 parts of the plurality of polyester fibers based on 100 parts of the composite textile **10**. At this fiber loading quantity and amount, the shoelace **14** may have comparatively very high untie-resistance, comparatively high tensile strength, comparatively low elasticity, comparatively high flame-resistance, a comparatively light weight, and a median melting temperature of about 150° C.

[0052] In another example of shoelaces **14** useful for lacing running footwear, the shoelace **14** may have a width **48** of 7 mm and may include 5 parts of the plurality of meta-aramid fibers and 95 parts of the plurality of polyester fibers based on 100 parts of the composite textile **10**. As this fiber loading quantity and amount, the shoelace **14** may have comparatively very high untie-resistance, comparatively high tensile strength, comparatively very high elasticity, comparatively low flame-resistance, a comparatively light weight, and a median melting temperature of about 90° C.

[0053] For shoelaces **14** useful for lacing track footwear, the shoelace **14** may have a width **48** of 6 mm and may include 70 parts of the plurality of meta-aramid fibers and 30 parts of the plurality of polyester fibers based on 100 parts of the composite textile **10**. At this fiber loading quantity and amount, the shoelace **14** may have comparatively very high untie-resistance, comparatively very high tensile strength, comparatively very low elasticity, comparatively moderate flame-resistance, a comparatively very light weight, and a median melting temperature of about 75° C.

[0054] For shoelaces **14** useful for lacing industrial footwear, the shoelace **14** may be round, may have a width **48** of 6 mm, and may include 85 parts of the plurality of meta-aramid fibers and 15 parts of the plurality of polyester fibers based on 100 parts of the composite textile **10**. At this fiber loading quantity and amount, the shoelace **14** may have comparatively very high untie-resistance, comparatively very high tensile strength, comparatively moderate elasticity, comparatively very high flame-resistance, a comparatively moderately heavy weight, and a median melting temperature of about 480° C. That is, the shoelace **14** may be thermally stable.

[0055] For shoelaces **14** useful for lacing hockey footwear, the shoelace **14** may have a width **48** of 15 mm and may include 30 parts of the plurality of meta-aramid fibers and 70 parts of the plurality of polyester fibers based on 100 parts of the composite textile **10**. At this fiber loading quantity and amount, the shoelace **14** may have comparatively very high untie-resistance, comparatively moderately high tensile strength, comparatively low elasticity, comparatively high flame-resistance, a comparatively light weight, and a median melting temperature of about 175° C.

[0056] For laces **14** useful as ankle ties for rowing, the lace **14** may have a width **48** of 3 mm and may include 85 parts of the plurality of meta-aramid fibers and 15 parts of the plurality of polyester fibers based on 100 parts of the composite textile **10**. At this fiber loading quantity and amount, the lace **14** may have comparatively very high untie-resistance, comparatively very high tensile strength, comparatively moderate elasticity, comparatively low flame-resistance, a comparatively very light weight, and a median melting temperature of about 40° C.

[0057] Tables 1-3 summarize the aforementioned properties.

TABLE 1

FOOTWEAR	WIDTH (mm)	ARAMID %	POLYESTER %	UNTIE-RESISTANCE	TENSILE STRENGTH
Hockey	15	30	70	very high	moderately high
Football (Running)	7	40	60	very high	high
Running	7	40	60	very high	high
Running	7	5	95	very high	high
Basketball	round/6	55	45	moderately high	high
Track	6	70	30	very high	very high
Tennis	8	80	20	very high	very high
Soccer	6	80	20	very high	very high
Football (Kicking)	6	80	20	very high	very high
Industrial	round/6	85	15	very high	very high
Rowing	3	85	15	very high	very high

TABLE 2

FOOTWEAR	WIDTH (mm)	ARAMID %	POLYESTER %	ELASTICITY	FLAME-RESISTANCE
Hockey	15	30	70	low	high
Football (Running)	7	40	60	low	high
Running	7	40	60	low	high
Running	7	5	95	very high	low
Basketball	round/6	55	45	high	high
Track	6	70	30	very low	moderate
Tennis	8	80	20	low	low
Soccer	6	80	20	high	moderate
Football (Kicking)	6	80	20	high	moderate
Industrial	round/6	85	15	moderate	very high
Rowing	3	85	15	moderate	low

TABLE 3

FOOTWEAR	WIDTH (mm)	ARAMID %	POLYESTER %	WEIGHT	MEDIAN MELTING TEMP. (° C.)
Hockey	15	30	70	light	175
Football (Running)	7	40	60	light	150
Running	7	40	60	light	150
Running	7	5	95	light	90
Basketball	round/6	55	45	light	110
Track	6	70	30	very light	75
Tennis	8	80	20	moderately heavy	50

TABLE 3-continued

FOOTWEAR	WIDTH (mm)	ARAMID %	POLYESTER %	WEIGHT	MEDIAN MELTING TEMP. (° C.)
Soccer	6	80	20	very light	50
Football (Kicking)	6	80	20	very light	50
Industrial	round/6	85	15	moderately heavy	480
Rowing	3	85	15	very light	40

[0058] Referring to Tables 1-3, when comparing the data for shoelaces **14** useful for lacing basketball and industrial footwear, for a given round shape and width **48** of 6 mm, as the quantity of the plurality of meta-aramid fibers substantially increases in the composite textile **10**, i.e., from 55 parts to 85 parts, untie-resistance, tensile strength, flame-resistance, weight, and median melting temperature increase. However, elasticity decreases.

[0059] Similarly, and again referring to Tables 1-3, when comparing the data for shoelaces **14** useful for lacing track and American football (kicking) footwear, for a given width **48** of 6 mm, as the quantity of the plurality of meta-aramid fibers increases slightly in the composite textile **10**, i.e., from 70 parts to 80 parts, elasticity substantially increases and the median melting temperature decreases.

[0060] With continued reference to Tables 1-3, when comparing the data for shoelaces **14** useful for lacing tennis and soccer footwear, for a given quantity of the plurality of meta-aramid fibers in the composite textile **10**, i.e., 80 parts, as the width **48** decreases from 8 mm to 6 mm, elasticity and flame-resistance increase and weight substantially decreases.

[0061] In addition, when comparing the data for shoelaces **14** useful for lacing footwear for running, for a given width **48** of shoelace **14**, as the quantity of the plurality of meta-aramid fibers in the composite textile **10** increases from 5 parts to 40 parts, elasticity substantially decreases, flame-resistance increases, and the median melting temperature increases. Stated differently, for a given width **48** of shoelace **14**, as the amount of the plurality of polyester fibers in the composite textile **10** increases from 60 parts to 95 parts, elasticity substantially increases, flame-resistance decreases, and the median melting temperature decreases.

Configurations of the Composite Textile

[0062] Referring now to FIGS. **4-7**, for certain shoelaces **14** or other applications, e.g., fabric **12** (FIG. **3**), the first yarn **20** and the second yarn **22** may be arranged in distinct configurations according to desired mechanical properties, performance, and aesthetics. Referring to FIG. **4**, in one non-limiting configuration, the first layer **16** and the second layer **26** may include the first yarn **20** and the second yarn **22** arranged in a staggered pattern **50** along an axis **52** that is oblique to the longitudinal axis **24**. That is, the staggered pattern **50** may form a series of alternating diagonal hashes or stripes on the shoelace **14**. Such alternating yarns **20**, **22** staggered along the oblique axis **52** may enhance the frictional interaction between the first yarn **20** and the second yarn **22** and between the first layer **16** and the second layer **26** to thereby limit translation along the longitudinal axis **24**. Further, although shown with a ratio or proportion of more first yarn **20** than second yarn **22** in FIG. **4**, the shoelace **14** may alternatively include more second yarn **22** than first yarn **20** or an equal amount of first and second yarn **20**, **22**.

[0063] Referring to FIG. **5**, in another non-limiting configuration, the first layer **16** and the second layer **26** may include the first yarn **20** and the second yarn **22** arranged in another staggered pattern **150** along the axis **52** that is oblique to the longitudinal axis **24**. That is, the staggered pattern **150** may form a series of alternating diagonal dots or hash marks on the shoelace **14**. Such alternating yarns **20**, **22** staggered along the oblique axis **52** may enhance the frictional interaction between the first yarn **20** and the second yarn **22** and between the first layer **16** and the second layer **26** to thereby limit translation along the longitudinal axis **24**. Further, although shown with a ratio or proportion of more second yarn **22** than first yarn **20** in FIG. **5**, the shoelace **14** may alternatively include more first yarn **20** than second yarn **22** or an equal amount of first and second yarn **20**, **22**.

[0064] Referring to FIG. **6**, in yet another non-limiting configuration, the first layer **16** and the second layer **26** may include the first yarn **20** and the second yarn **22** arranged in yet another staggered pattern **250** along the axis **52** that is oblique to the longitudinal axis **24**. That is, the staggered pattern **250** may form a series of alternating diagonal or vertical stripes on the shoelace **14**. Such alternating yarns **20**, **22** staggered along the oblique axis **52** may enhance the frictional interaction between the first yarn **20** and the second yarn **22** and between the first layer **16** and the second layer **26** to thereby limit translation along the longitudinal axis **24**. Further, although shown with a ratio or proportion of equal amounts of first yarn **20** and second yarn **22** in FIG. **6**, the shoelace **14** may alternatively include more first yarn **20** than second yarn **22** or more second yarn **22** than first yarn **20**.

[0065] Referring to FIG. **7**, in a further non-limiting configuration, the first layer **16** and the second layer **26** may include the first yarn **20** arranged in a first zig-zag pattern **54** along a first horizontal axis **56** that is perpendicular to the longitudinal axis **24**. In addition, the first layer **16** and the second layer **26** may include the second yarn **22** arranged in a second zig-zag pattern **58** that abuts the first zig-zag pattern **54** along a second horizontal axis **60** that is parallel to the first horizontal axis **56**. That is, the shoelace **14** may appear to have alternating zig-zag patterns **54**, **58**. Such alternating yarns **20**, **22** arranged in the zig-zag patterns **54**, **58** may enhance the frictional interaction between the first yarn **20** and the second yarn **22** and between the first layer **16** and the second layer **26** to thereby limit translation along the longitudinal axis **24** in response to the tensile force **34** (FIG. **2**).

[0066] Referring again to FIG. **1**, the composite textile **10** may further include at least one additional layer **62** formed from the two-yarn composite **18** and abutting the second layer **26** along the longitudinal axis **24** to therefore sandwich the second layer **26** between the first layer **16** and the at least one additional layer **62**. For example, the composite textile **10** may include a third layer **62** formed from the two-yarn composite **18** and abutting the second layer **26** along the longitudinal axis **24** to thereby sandwich the second layer **26** between the first layer **16** and the second layer **26**. In some examples, the composite textile **10** may have four or five or six layers **62**.

[0067] Therefore, in summary, due to both a) the interaction of the plurality of hooks **32**, the first plurality of micro-ridges **38**, and the second plurality of micro-ridges **40** to create frictional resistance to translation of the first layer **16** with respect to the second layer **26** along the longitudinal axis **24** in response to the tensile force **34** and b) the ratio of the plurality of meta-aramid fibers to the plurality of polyester fibers, the composite textile **10** may have excellent tensile strength, near-perfect elasticity, high grip, superior untie-resistance, and excellent flame retardance. As such, articles such as shoelaces **14** formed from the composite textile **10** are resistant to becoming untied during athletic activity and provide excellent comfort to a user. Other articles formed from the composite textile **10** such as athletic recovery equipment formed from fabric **12**, e.g., athletic tapes and ankle braces, exhibit excellent strength, elasticity, and durability, enhance athletic performance, and aid in muscular recovery.

[0068] The described embodiments of the present disclosure are intended to serve as non-limiting examples, and other embodiments may take various and alternative forms. In addition, the appended drawings are not necessarily to scale, and may present a somewhat simplified representation of various features of the present disclosure, including, for example, specific dimensions, orientations, locations, and shapes. Details associated with such features will be determined in part by the intended application and use environment of the described embodiments.

[0069] For purposes of the present description, unless specifically disclaimed, use of the singular includes the plural and vice versa, the terms "and" and "or" shall be both conjunctive and disjunctive, and the words "including", "containing", "comprising", "having", and the like shall mean "including without limitation". Moreover, words of approximation such as "about", "substantially", "generally", "approximately", etc., may be used herein in the sense of "at, near, or nearly at", or "within 0-5% of", or "within acceptable manufacturing tolerances", or logical combinations thereof. As used herein, a component that is "configured to" perform a specified function is capable of performing the specified function without alteration, rather than merely having potential to perform the specified function after further modification. In other words, the described hardware, when expressly configured to perform the specified function, is specifically selected, created, implemented, utilized, programmed, and/or designed for the purpose of performing the specified function. In addition, the use of ordinals such as first, second and third does not necessarily imply a ranked sense of order, but rather may merely distinguish between multiple instances of an act or structure.

[0070] The detailed description and the drawings or figures are supportive and descriptive of the present teachings, but the scope of the present teachings is defined solely by the claims. While some of the best modes and other embodiments for carrying out the present teachings have been described in detail, various alternative designs and embodiments exist for practicing the present teachings defined in the appended claims. Moreover, this disclosure expressly includes combinations and sub-combinations of the elements and features presented above and below.

What is claimed is:

1. A composite textile comprising:

a first layer formed from a two-yarn composite that includes a first yarn formed from a plurality of meta-aramid fibers and a second yarn formed from a plurality of polyester fibers;

wherein the first layer extends along a longitudinal axis; and

a second layer formed from the two-yarn composite and abutting the first layer along the longitudinal axis;

wherein the first yarn has a first surface having a first surface finish;

wherein the second yarn has a second surface spaced opposite the first surface and having a second surface finish that is smoother than the first surface finish;

wherein the first yarn includes a plurality of hooks each extending from the first surface toward the second surface;

wherein the first layer includes a first plurality of micro-ridges disposed along the first surface and the second layer includes a second plurality of micro-ridges disposed along the second surface; and

wherein the first plurality of micro-ridges and the second plurality of micro-ridges interact to generate friction between the first layer and the second layer and limit movement of the first layer and the second layer with respect to one another as the composite textile extends along the longitudinal axis in response to a tensile force.

2. The composite textile of claim **1**, wherein the plurality of meta-aramid fibers are present in the composite textile in a quantity of from 30 parts to 95 parts based on 100 parts of the composite textile.

3. The composite textile of claim **2**, wherein the plurality of polyester fibers are present in the composite textile in an amount of from 5 parts to 70 parts based on 100 parts of the composite textile.

4. The composite textile of claim **1**, wherein the first layer and the second layer align along the longitudinal axis such that the first plurality of micro-ridges and the second plurality of micro-ridges contact one another when one of the first layer and the second layer moves along the longitudinal axis with respect to another of the first layer and the second layer in response to the tensile force.

5. The composite textile of claim **1**, wherein the plurality of hooks attach to the second surface along the longitudinal axis to limit movement of one of the first layer and the second layer with respect to another of the first layer and the second layer along the longitudinal axis in response to the tensile force.

6. The composite textile of claim **1**, wherein each of the plurality of hooks have a height of from 1 micron to 50 microns.

7. The composite textile of claim **1**, wherein the first yarn is braided with the second yarn about the longitudinal axis.

8. The composite textile of claim **1**, wherein the first yarn is woven with the second yarn about the longitudinal axis.

9. The composite textile of claim **1**, wherein the first yarn is knitted with the second yarn about the longitudinal axis.

10. The composite textile of claim 1, further including at least one additional layer formed from the two-yarn composite and abutting the second layer along the longitudinal axis to therefore sandwich the second layer between the first layer and the at least one additional layer.

11. The composite textile of claim 1,

wherein the first yarn has a yarn count of 21 meta-aramid fibers and a single ply; and

wherein the second yarn is a spun yarn.

12. A shoelace formed from the composite textile of claim 1 and including an aglet disposed at each end of the shoelace.

13. The shoelace of claim 12, wherein the shoelace has a width of from 3 millimeters (mm) to 30 mm.

14. A fabric formed from the composite textile of claim 1.

15. A composite textile comprising:

a first layer formed from a two-yarn composite that includes a first yarn formed from a plurality of meta-aramid fibers and a second yarn formed from a plurality of polyester fibers;

wherein the first layer extends along a longitudinal axis;

a second layer formed from the two-yarn composite and abutting the first layer along the longitudinal axis; and

a third layer formed from the two-yarn composite and abutting the second layer along the longitudinal axis to thereby sandwich the second layer between the first layer and the second layer;

wherein the plurality of meta-aramid fibers and the plurality of polyester fibers are present in the composite textile in a ratio of a quantity of the plurality of meta-aramid fibers to an amount of the plurality of polyester fibers of from 5:95 to 95:5;

wherein the first yarn has a first surface and the second yarn has a second surface spaced opposite the first surface;

wherein the first yarn includes a plurality of hooks each extending from the first surface toward the second surface;

wherein the first layer includes a first plurality of micro-ridges disposed along the first surface and the second layer includes a second plurality of micro-ridges disposed along the second surface; and

wherein the first plurality of micro-ridges and the second plurality of micro-ridges interact to generate friction between the first layer and the second layer and limit movement of the first layer and the second layer with respect to one another as the composite textile extends along the longitudinal axis in response to a tensile force.

16. The composite textile of claim 15, wherein the first layer and the second layer include the first yarn arranged in a first zig-zag pattern along a first horizontal axis that is perpendicular to the longitudinal axis; and

wherein the first layer and the second layer include the second yarn arranged in a second zig-zag pattern that abuts the first zig-zag pattern along a second horizontal axis that is parallel to the first horizontal axis.

17. The composite textile of claim 15, wherein the first layer and the second layer include the first yarn and the second yarn arranged in a staggered pattern along an axis that is oblique to the longitudinal axis.

18. The composite textile of claim 15, wherein the first layer and the second layer align along the longitudinal axis such that the first plurality of micro-ridges and the second plurality of micro-ridges contact one another when the first layer translates along the longitudinal axis with respect to the second layer in response to the tensile force.

19. The composite textile of claim 18, wherein the plurality of hooks attach to the second surface along the longitudinal axis to limit translation of the first layer with respect to the second layer along the longitudinal axis in response to the tensile force.

20. A composite textile comprising:

a first layer formed from a two-yarn composite that includes a first yarn formed from a plurality of meta-aramid fibers and a second yarn formed from a plurality of polyester fibers;

wherein the first layer extends along a longitudinal axis; and

a second layer formed from the two-yarn composite and abutting the first layer along the longitudinal axis;

wherein the plurality of meta-aramid fibers and the plurality of polyester fibers are present in the composite textile in a ratio of a quantity of the plurality of meta-aramid fibers to an amount of the plurality of polyester fibers of from 5:95 to 95:5;

wherein the first yarn has a first surface having a first surface finish;

wherein the first yarn has a yarn count of 21 meta-aramid fibers and a single ply;

wherein the second yarn is a spun yarn and has a second surface spaced opposite the first surface and having a second surface finish that is smoother than the first surface finish;

wherein the first yarn includes a plurality of hooks each extending from the first surface toward the second surface and having a height of from 1 micron to 50 microns;

wherein the first layer includes a first plurality of micro-ridges disposed along the first surface and the second layer includes a second plurality of micro-ridges disposed along the second surface;

wherein the first plurality of micro-ridges and the second plurality of micro-ridges contact one another to generate friction between the first layer and the second layer and limit movement of the first layer and the second layer with respect to one another as the composite textile extends along the longitudinal axis in response to a tensile force; and

wherein the plurality of hooks attach to the second surface along the longitudinal axis to limit movement of one of the first layer and the second layer with respect to another of the first layer and the second layer along the longitudinal axis in response to the tensile force.

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